Grid-Based Quantum Annealing for Sudoku

A QUBO Formulation of Sudoku as Graph Coloring — Benchmarked Against Classical Simulated Annealing

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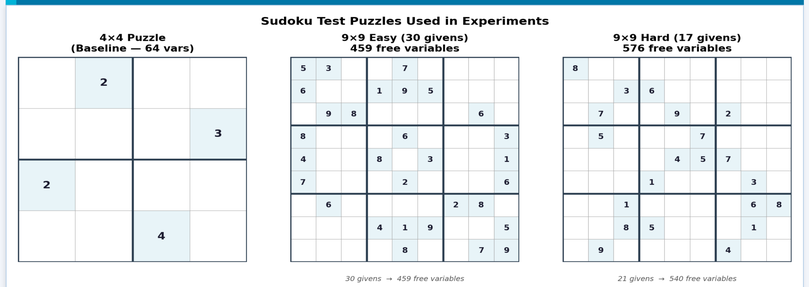
Background & Motivation

- Sudoku encodes constraint satisfaction structures in scheduling, logistics, and resource allocation — an ideal quantum benchmark.
- Quantum annealing uses tunneling to escape local energy minima, potentially outperforming classical heuristics.
- Goal: establish baselines for when quantum methods surpass

Sudoku Puzzles Used in Experiments



4×4 baseline (64 vars) · 9×9 Easy (459 free vars) · 9×9 Hard (576 free vars)

Problem Formulation

QUBO Encoding:

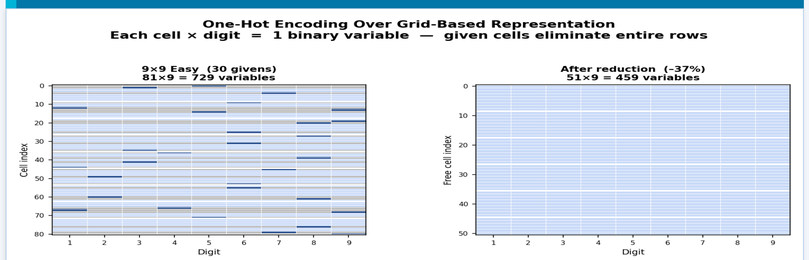
x(i,j,k) = 1 if cell (i,j) holds digit k+1, else 0

Four Constraint Penalties:

- E_1 Cell · E_2 Row · E_3 Column · E_4 Box

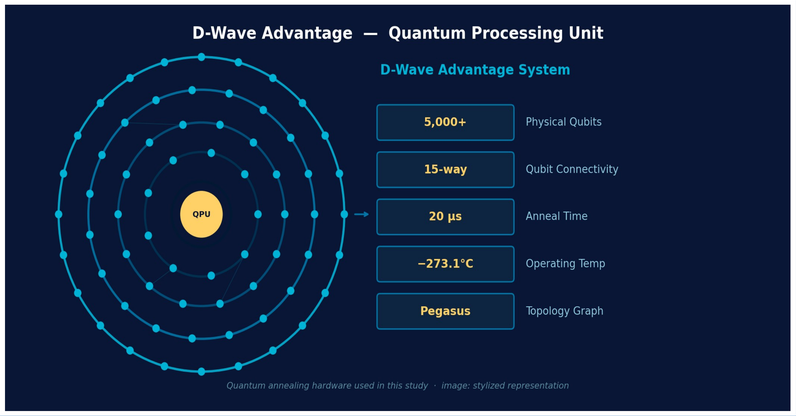
$E_{total} = E_1+E_2+E_3+E_4 = 0 \iff$ Valid Sudoku

One-Hot Encoding Over Grid-Based Representation



9×9 Easy: 729 variables before (left) vs 459 after given-cell elimination (right). Dark = fixed, blue = free.

Quantum Annealing Hardware



Methodology

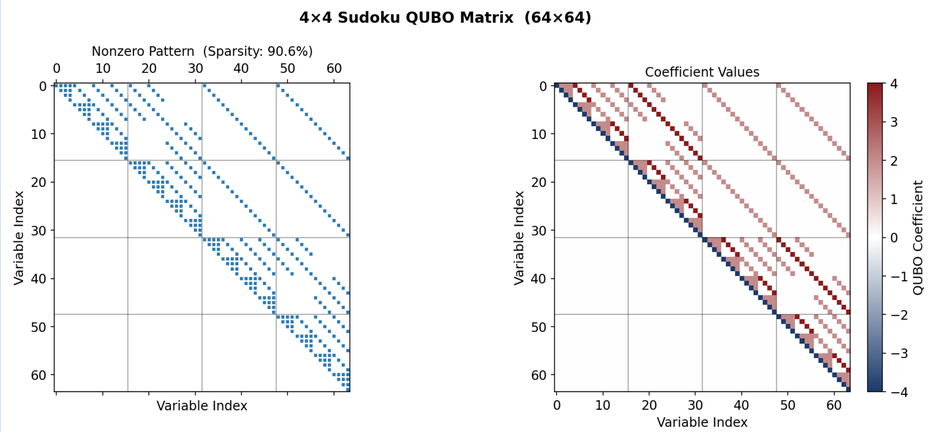
Solvers Evaluated

- Simulated Annealing (D-Wave neal)
- QAOA (Qiskit) — 2×2 only
- Simulated Bifurcation — 4×4 only
- QA.jl (Julia) — infeasible >20 qubits

Parameters

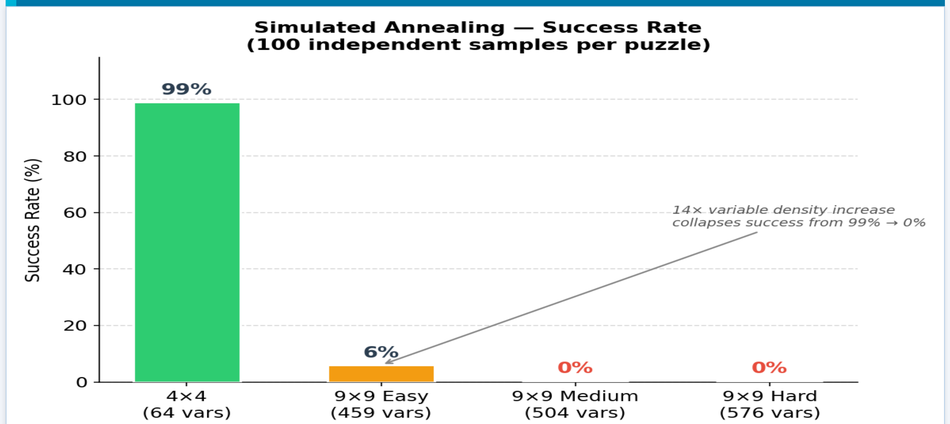
- SA: 100 reads × 1000 sweeps
- QAOA: p=2, 2048 shots, COBYLA
- All λ weights = 1.0
- Metrics: success rate, best energy

4×4 QUBO Matrix Structure



Sparsity pattern (left, 90.6% zeros) and coefficient values (right). Block structure encodes the four constraint types.

Simulated Annealing — Success Rate



14× variable density increase (64→459 vars) collapses success from 99% to 0%.

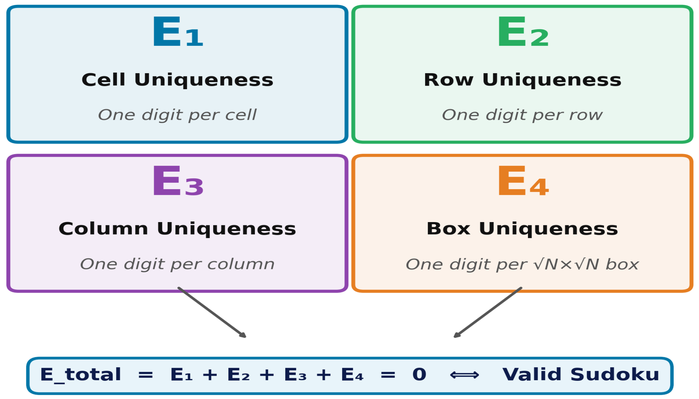
D-Wave Hybrid Solver vs. Simulated Annealing

Puzzle	Vars	SA	SA E	Hybrid E	Time
Easy (30 clues)	459	6% ✓	0 ✓	0 ✓	1.64s
Medium (25 clues)	504	0% ✗	4 ✗	0 ✓	1.80s
Hard (17 clues)	576	0% ✗	4 ✗	4 ✗	1.72s

★ Quantum Tunneling Advantage Demonstrated

Hybrid solver succeeded on Medium where all 100 SA runs failed — tunneling past classical local minima.

Constraint Encoding



QAOA Results

- 2×2 Sudoku (8 qubits): energy 4.0, but post-processing recovered a valid solution.
- 4×4 impossible: 2^{64} states ≈ 148 exabytes of RAM.
- NISQ noise and exponential state space block practical scaling.

Why Quantum Annealing Fits

- D-Wave natively solves QUBOs — 5,000+ qubits, no circuit depth limits.
- Sudoku QUBOs are 84–97% sparse — favorable for hardware embedding.
- Hybrid solver pairs classical preprocessing with quantum tunneling.

Conclusions

- ✓ QUBO correctly encodes all constraints — E=0 ↔ valid solution.
- ✓ SA: 99% on 4×4, but 6% / 0% / 0% on 9×9 easy / medium / hard.
- ✓ D-Wave hybrid solved 2/3 puzzles vs SA's 1/3 — quantum advantage on medium.
- ✗ Both solvers failed on 17-clue Sudoku — hardest instances remain open.
- → Next: D-Wave Advantage QPU, penalty tuning, domain-wall encoding.

References & Acknowledgements

Advisor: Dr. Thang Dinh, VCU

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